Practical # 11

Objective:

Implement Selection, Bubble, and Insertion sort; analyze their time complexities.

Theory:

Sorting refers to the operation or technique of arranging and rearranging sets of data in some specific order.

Selection Sort works:

The selection sort algorithm is performed using the following steps:
Step 1 - Select the first element of the list (i.e., Element at first position in the list).
Step 2: Compare the selected element with all the other elements in the list.
Step 3: In every comparison, if any element is found smaller than the selected element (for Ascending order), then both are swapped.
Step 4: Repeat the same procedure with element in the next position in the list till the entire list is sorted.

Bubble Sort work:

The Bubble sort algorithm is performed using the following steps:
Step 1 - Bubble sort starts with very first two elements, comparing them to check which one is greater. Put larger one at higher index.
Step 2 - It takes next two values compare these values and place larger one at higher index.
Step 3 - This process does iteratively until the largest value is not reached at the last index. Then start again from 0 (zero) index up to n-1 index.
Step 4 - The algorithm follows the same steps iteratively until elements are not sorted

Insertion Sort work:

The insertion sort algorithm is performed using the following steps:
Step 1 - Assume that first element in the list is in sorted portion and all the remaining elements are in unsorted portion.
Step 2: Take first element from the unsorted portion and insert that element into the sorted portion in the order specified.
Step 3: Repeat the above process until all the elements from the unsorted portion are moved into the sorted portion.

In this lab, students will learn how to:

- Implement Selection, Bubble, and Insertion sorting algorithms in C++.
- Apply sorting algorithms to organize data in ascending and descending order.
- Analyze the performance (number of comparisons and swaps) of each algorithm.
- Evaluate the time complexity ($O(n^2)$, $O(n)$, etc.) in different cases.
- Compare the efficiency of Selection, Bubble, and Insertion sort for small vs. large datasets.

C++ program: Write C++ program to arrange elements in ascending order using selection sorting algorithm.

```cpp
#include<iostream>
using namespace std;
int main()
{
    int tot, arr[50], i, j, temp, small, chk, index;
    cout<<"Enter the Size of Array: ";
    cin>>tot;
    cout<<"Enter "<<tot<<" Array Elements: ";
    for(i=0; i<tot; i++)
        cin>>arr[i];
    for(i=0; i<(tot-1); i++)
    {
        chk=0;
        small = arr[i];
        for(j=(i+1); j<tot; j++)
        {
            if(small>arr[j])
            {
                small = arr[j];
                chk++;
                index = j;
            }
        }
        if(chk!=0)
        {
            temp = arr[i];
            arr[i] = small;
            arr[index] = temp;
        }
    }
    cout<<"\nSorted Array is:\n";

    for(i=0; i<tot; i++)
        cout<<arr[i]<<" ";
    cout<<endl;
    return 0;
}
```

OUTPUT

```
Enter the Size of Array: 4
Enter 4 Array Elements: 45
25
87
31

Sorted Array is:
25 31 45 87
```

Review Questions/ Exercise:

1. Write a C++ program to implement **Selection Sort** and count the number of comparisons.
2. Write a C++ program to implement **Bubble Sort** and print the array after each pass.
3. Write a C++ program to implement **Insertion Sort** and display intermediate steps.
4. Implement all three algorithms for the same dataset and record the execution time.
5. Construct a program to sort elements in **descending order** using Bubble Sort.

Name: ______________________________

Roll #: ______________________________

Date: ______________________________

Subject Teacher